Stablecoin StressBench: The Highest-AUROC Model Loses the Most Money Under a 12× Execution Gap That Vanishes On-Chain

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Abstract

On 10 March 2023, Silicon Valley Bank's seizure forced USDC off its dollar peg. Price charts showed 34.3% of one-minute windows as profitable stablecoin arbitrage; fewer than 3 in 100 (2.88%) survived VWAP book-walking, fees, and settlement, a 12× **gap** between visible and executable profit (proxy-bounded to 12–14× under a conservative 20% depth haircut). A model scoring well on price labels can still lose money on every trade. We release **Stablecoin StressBench**, an 18-event catalogue across seven mechanism classes with per-minute VWAP (volume-weighted average price) book-walking labels, route provenance, and a hindsight-oracle ceiling (CC-BY 4.0), and use it to ask what works and what does not. *What does not work:* every calm-trained family we test (tabular, sequence, reinforcement-learning, and change-point) loses money despite high AUROC; the highest-AUROC model (GRU, 0.80) is the largest loser (−239 bps), a direct quantification of the AUROC–P&L inversion under class imbalance. A pre-registered 81-path search over training protocols, feature sets, and models finds **zero** paths that select profitable windows out-of-sample under a Benjamini–Hochberg FDR(0.10) gate; even purged in-event cross-validation (train and test *within* the SVB crisis) is strongly negative (−66 to −122 bps), so executable-window *selection* is unsolved, not merely non-transferable. *What does work:* the barrier is venue-specific. On real on-chain Curve/AMM data, where execution cost is the bounded pool slippage rather than order-book impact, a simple deviation rule is profitable in 5/5 episodes; 4/5 survive a conservative 30 bps gas haircut (+169 to +463 bps, moving-block bootstrap 95% CIs above zero), the lone exception being the small USDC/SVB on-chain dislocation. The same signal that loses money on every CEX path is capturable on-chain, so the optical-to-executable barrier is a property of CEX order-book microstructure, not of stablecoin depegs.

Keywords

transfer learning, meta-labeling, reinforcement learning, execution-aware labels, market microstructure, stablecoin arbitrage, financial benchmarks

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1 Introduction

Stablecoin stress is not one phenomenon but seven. A *stablecoin* is a cryptocurrency engineered to hold a fixed value—almost always one US dollar—and serves as the dollar settlement rail of crypto markets; when it loses that anchor (a *depeg*), the event is at once a systemic-risk shock and, on the price chart, an apparent arbitrage. From 2020 to 2023 we catalogue 18 distinct depeg episodes spanning algorithmic death spirals (Terra/LUNA, IRON/TITAN), fiat-reserve shocks (USDC/SVB), regulatory winddowns (BUSD), exchange-credit collapses (FTX, Celsius), DeFi pool imbalances (Curve/USDT), collateral liquidations (DAI Black Thursday), and real-world-asset failures (USDR, Acala). Yet every prior study examines a single mechanism, and none asks the question a trader actually faces. Of the price dislocations that appear during stress, how many can be captured at a profit?

The answer is far fewer than the chart suggests. When USDC slipped below $1 during the March 2023 SVB crisis, Bitcoin bought with USDC looked cheaper than the same Bitcoin priced in real dollars, free money on a price chart. In practice it rarely was. By the time an order walked the book, the discount had narrowed. Taker fees and settlement delay consumed the rest. Only 2.88% of one-minute windows survived all three costs, against 34.3% that appeared dislocated, a 12× gap between visible and executable arbitrage.

We separate three layers of the same event. **Layer 1 (optical)** is what the price chart shows. A stablecoin's *basis*, its deviation from $1, appears positive, suggesting a profitable arbitrage. **Layer 2 (executable)** is what survives real execution. A market order does not fill at the quoted price. It consumes successive price levels in the order book, and the resulting VWAP (volume-weighted average price, $\bar{P} = \sum_i P_i Q_i / \sum_i Q_i$ across consumed levels) worsens with size. Layer 2 keeps only the minutes where the VWAP-filled profit, after fees and settlement delay, is still positive. **Layer 3 (captured)** is what a model can flag in advance. Gap 1 (Layer 1⟶2) is a measurement problem: price labels overstate tradeable performance by 12×. Gap 2 (Layer 2⟶3) is a training problem: calm-trained models fail on unseen stress regimes. Layer-1-only dislocations are ***optical***; survivors are ***executable***. Figure 2 shows all three layers for the SVB test window and traces one representative false-positive minute from apparent profit to net loss. Basis points (bps) are hundredths of a percent (100 bps = 1%).

We establish each gap against a ground truth. Terra/LUNA, a mechanistically independent algorithmic collapse, is our known reference (gap 5.9×, 2.40% executable) and supplies the cross-mechanism training signal; USDC/SVB, a fiat-reserve shock with route-complete order-book depth, is the execution-grade test case (gap 12×, 2.88% executable, widening to 14× under a 20% depth haircut). The same gap in two unrelated mechanisms makes the phenomenon structural, not an artefact of either event; only the *Tier-A* execution labels, backed by real full-depth order-book data rather than a price-only proxy, are SVB-specific.

No calm-trained model captures this gap. In stress the order book withdraws depth and widens spreads, but *which* dislocated minutes are executable turns out to be unpredictable from that microstructure—we show below that even a model trained and

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tested *within* the SVB crisis cannot select them. The barrier is not the crisis, it is the venue: on an automated market maker the same dislocation is capturable, because execution cost is the bounded slippage of the pool invariant rather than open-ended order-book impact.

Contributions.

(1) **An execution-aware benchmark that exposes a 12× optical-to-executable gap.** StressBench supplies the apparatus: an 18-event source-tracked catalogue across seven mechanism classes, per-minute VWAP book-walking labels with route provenance, and a *hindsight-oracle ceiling* that upper-bounds any model's profit. On the marquee USDC/SVB event, 34.3% of minutes look dislocated but only 2.88% are executable (a 12× gap), so oracle capture—not price-label AUROC—is the right yardstick. (§3–5)

(2) **What does not work: executable-window selection is unsolved on the CEX order book.** Every calm-trained family loses money despite high AUROC, and the AUROC–P&L relationship *inverts* (the highest-AUROC model, GRU 0.80, loses −239 bps). A pre-registered 81-path search (training protocols × feature sets × models) yields zero paths that clear a Benjamini–Hochberg FDR(0.10) gate [23]; even purged in-event cross-validation is strongly negative, so the windows are unpredictable from this microstructure, not merely non-transferable. (§6.3, 6.4)

(3) **What does work: the gap is venue-specific and the on-chain venue is tradeable.** On real on-chain Curve/AMM data, a simple deviation rule is profitable in 5/5 episodes and 4/5 survive a conservative 30 bps gas haircut (+169 to +463 bps, moving-block bootstrap CIs > 0). The optical-to-executable barrier is therefore a property of CEX order-book microstructure—thin depth, taker fees, settlement delay—not of stablecoin depegs. (§6.2, 6.4)

What you can do with this benchmark (CC-BY 4.0). The full catalogue, labels, oracle, and harness are released under CC-BY 4.0: test any model against 18 real depeg episodes without data collection, measure oracle capture as a principled denominator, and reproduce the 12× gap from provided scripts.

Figure 1 summarises the full benchmark pipeline, from the 18-event catalogue through execution labels and the model ladder to the released artefacts.

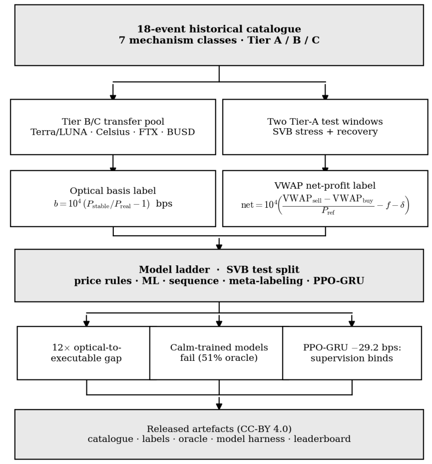

Figure 1: StressBench pipeline. The 18-event catalogue is filtered to Tier-A execution-grade windows for VWAP labels and oracle evaluation; Tier-B/C events supply mechanism-diverse transfer training. Models are ranked by mean net bps per triggered trade, not price-label AUROC.

2 Related Work

Stablecoin stress and depeg detection. Uhlig [1] and Griffin and Shams [2] analyse Terra/LUNA and stablecoin non-fundamental demand. Hernandez Cruz et al. [20] study DEX liquidity during the SVB collapse without modelling CEX execution costs. Cintra and Holloway [9] detect the Terra depeg 12 hours early via BOCPD on AMM reserves. We replicate their approach on CEX cross-quote data and find AUROC 0.229, confirming that AMM and CEX signals differ fundamentally. None of these works provide VWAP-grade execution labels, a hindsight oracle, or cross-mechanism transfer evaluation. Our design therefore follows three principles from the broader literature. Cont and Kukanov [6] show VWAP book-walking is the correct profitability measure for sequential execution, we implement this exactly. Shleifer and Vishny [4] show that visible arbitrage gaps need not survive execution frictions, our 12× gap is a direct measurement of this effect at one-minute resolution. And Lopez de Prado [8] shows that a meta-labeler trained on the execution outcome outperforms direction predictors, our cross-mechanism result is the evidence that this extends across stress mechanisms.

Limits to arbitrage and execution frictions. Shleifer and Vishny [4], Gromb and Vayanos [5], and Makarov and Schoar [7] show that visible price gaps need not be profitable once execution frictions are included. Our 12× ratio is this effect measured at one-minute resolution. Hautsch et al. [3] quantify settlement latency as a first-order cost on blockchain assets. Cont and Kukanov [6] establish VWAP book-walking as the correct profitability measure. Gogol et al. [18] measure Maximal Arbitrage Value on AMM-to-CEX routes. Their partial execution labels cover AMM venues only and lack a hindsight oracle. Adams et al. [19] study swap costs on Uniswap without stress-event execution analysis.

ML and RL for execution. Lopez de Prado [8] introduces *meta-labeling*. A two-stage architecture in which a primary model (here, the Layer-1 dislocation flag) proposes candidate trades, and a secondary *meta-labeler* learns which of those proposals will actually be profitable. The meta-labeler thus predicts the binary label "this candidate survives execution" rather than predicting the market direction from scratch. We extend this to the *cross-mechanism* setting, where the meta-labeler is trained on one crisis type (Terra/LUNA) and applied unchanged to another (USDC/SVB). Moallemi and Wang [16] and Ning et al. [17] supply the finite-horizon MDP framing we adopt for the RL baseline. Mohl et al. [21] and Olby et al. [22] operate in reactive LOB environments with market-impact

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	partial/proxy, real L2 where archived	real L2 (futures)
Sell BTC-USDT	raw/futures depth	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024. All 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 one-minute bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

feedback. Our diagnostic uses historical replay without feedback, isolating training supervision as the variable.

3 Benchmark Design

3.1 Data Sources and Instruments

The 18-event catalogue defines the historical stress universe. The six-window empirical panel (56,134 one-minute bars) defines the model dataset, and the Tier-A SVB windows define the execution-grade testbed. Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs across two centralised exchanges (CEXs, i.e. Binance and Coinbase). The legs are Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference). The BTC-USDC buy leg uses kline-proxy depth for the SVB window because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness (gap widens to 14×, oracle stays above +130 bps).

Data quality varies by leg, and every record carries a provenance tag (Table 1).

3.2 Event-Based Split Design

Splits are defined by event identity, so no model trained on the calm split has encountered stress dynamics. Threshold calibration uses the Terra/LUNA validation split, an independent stress event, so the SVB test split is never observed during model development. The 2.88% positive rate in the test split (Table 2) implies severe class imbalance. We therefore report economic net bps as the primary criterion. AUROC and AUPRC are secondary metrics.

Five leakage checks confirm validity. Label shuffles produce negative P&L. Lookahead does not improve performance. Train and test are separated by event identity.

The full 18-event catalogue with per-event mechanism taxonomy and Tier classifications is available in the project repository.

3.3 Benchmark Positioning

No prior benchmark (§2) combines per-minute executable CEX labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation; StressBench provides all four.

Feature sets. Four nested sets (`price_only` ⊂ `price_plus_book` ⊂ `price_book_frag` ⊂ `price_book_settle`) isolate marginal signal from price basis, microstructure (spread, depth, imbalance), fragmentation, and on-chain settlement proxies (kline-proxy for SVB/Terra windows, Tier B).

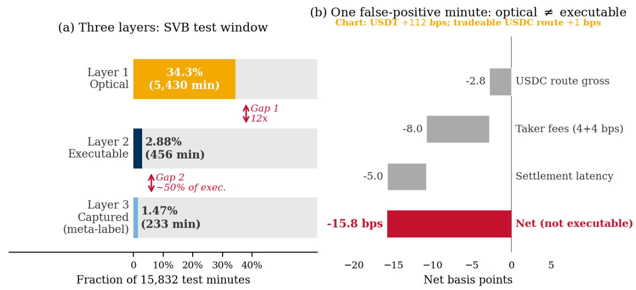

Figure 2: (a) Three-layer framework for the SVB test window (15,832 min). Layer 1 optical (34.3%), Layer 2 executable (2.88%), Layer 3 captured by cross-mechanism meta-labeling (1.47%). Gap 1 (12×) is a measurement problem. Gap 2 is a training problem. (b) Worked example. A minute that appears profitable on the chart nets −15.8 bps after VWAP book-walking, fees, and route-direction mismatch.

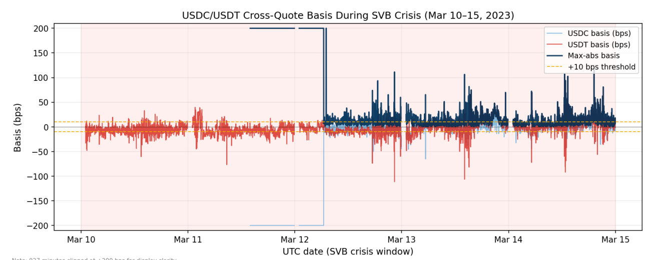

Figure 3: Cross-quote basis during the SVB test window (Mar 10–20 2023). USDC basis (navy) exceeds 10 bps in 12.4% of minutes, USDT basis (red) in 27.4%, and max-abs (gold) in 34.3%. The price-to-execution ratio is 12× (0% depth haircut; 14× under a 20% haircut).

4 Execution-Aware Labels

4.1 Three-Layer Framework

Figure 2 structures the paper's central argument visually, instantiating the three layers and two gaps defined in §1: both gaps are nonzero, independent, and large. Panel (b) traces one representative false-positive minute through the full cost stack, showing why 34.3% optical activity collapses to 2.88% executable.

4.2 Cross-Quote Basis and Labels

The cross-quote basis is the percentage gap between BTC priced in a stablecoin (e.g. BTC-USDC) and BTC priced in real dollars (BTC-USD on Coinbase). A non-zero basis signals an apparent dislocation (Figure 3 shows it across the SVB window). Three equations define the label construction.

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad [\mathrm{bps}] \qquad (1)$$

$$b_{\max}(t) = \max(|b_{\mathrm{USDC}}(t)|,\ |b_{\mathrm{USDT}}(t)|) \qquad (2)$$

$$y_{\mathrm{basis}}(t) = \mathbf{1}[|b_{\mathrm{USDC}}(t+h)| > \tau] \qquad (3)$$

Here $\mathbf{1}[\cdot]$ is the indicator function; the primary task is `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.3 VWAP Execution and Net Profit Labels

$$\text{net}(q,t) \;=\; 10{,}000\times\left[\frac{\text{VWAP}_{\text{sell}}-\text{VWAP}_{\text{buy}}}{P_{\text{ref}}}-f_{\text{buy}}-f_{\text{sell}}-\delta_{\text{settle}}\right] \tag{4}$$

$$y_{\text{arb}}(q,h,t) \;=\; \mathbf{1}\left[\max_{s\in[t,\,t+h]}\text{net}(q,s) \;>\; c\right] \tag{5}$$

Profit threshold c=10 bps, taker fee 4 bps per side, and δ=5 bps. The δ assumption is validated against 100K on-chain USDC ERC-20 transfers. Median cost was 1.4–3.2 bps on nine of ten SVB days. The one exception (Mar 11 gas spike) saw P95 reach 8.6 bps.

$$\text{Oracle}(q) \;=\; \frac{1}{|\mathcal{T}^+|}\sum_{t\in\mathcal{T}^+}\text{net}(q,t), \qquad \mathcal{T}^+=\{t:\text{net}(q,t)>c\} \tag{6}$$

The oracle is a hypothetical strategy that trades every profitable window with perfect hindsight and represents an upper bound no deployable model can exceed. Oracle capture $\kappa_M=\bar{r}_M/\text{Oracle}(q)$ is the fraction of the oracle's per-trade return recovered by model M.

5 Models and Evaluation

5.1 Model Ladder

The ladder is a sequential elimination experiment with one result at every rung. Every family fails for the same reason, distributional mismatch between calm training and stress test.

Architecture. No help. The GRU posts the highest AUROC among sequence models (0.80) and the worst economic return (−239 bps, Figure 4). Discriminability does not translate to profit at 2.88% positive rate.

Feature engineering. No help. All four nested feature sets produce negative returns. Depth and settlement signals offer no lift when the training split contains zero stress-regime examples.

Class balancing. No help. Cost-sensitive reweighting cannot synthesise stress dynamics from calm data. The issue is distributional, not proportional.

Regression targeting. No help. ExpNetProfitRegressor optimises the economic criterion directly and reaches −73.8 bps. Label mismatch is not the explanation.

Regime detection. No help. Bayesian Online Change Point Detection (BOCPD), CUSUM, and EWMA all fail: CEX basis is mean-reverting, not change-point structured (AUROC 0.229).

RL reward optimisation. No help at identical density. A conditioned PPO-GRU at 17% primary-signal density earns −29.2 bps. The reward signal is not a substitute for explicit labels even when positive-example density is matched.

The cross-mechanism extension of meta-labeling operates in three steps. Meta-labeling [8] is a two-step filter. A cheap rule flags candidate minutes, then a learned classifier keeps only the ones likely to survive execution.

(1) **Primary filter.** Flag t if $|b_{\text{USDC}}(t)|>\tau_1$ (τ_1=10 bps).
(2) **Secondary classifier.** Fit f_θ on $\{(x_t,\,y_{\text{arb}}(t))\}_{t\in\mathcal{E}_{\text{train}}}$, where $\mathcal{E}_{\text{train}}$ is a mechanistically distinct stress event.
(3) **Entry.** Signal at t iff primary fires and $f_\theta(x_t)>\theta^*$, where $\theta^*=\arg\max_\theta\sum_{i:\hat{p}_i>\theta}\text{net}(q,t_i)$ on the Terra/LUNA calibration block.

Table 3: Master oracle and gap table. Every oracle value cited in the paper appears here. Gap = oracle minus best deployable model.

Task / Window	N	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m` (full test)	15,832	+161.7 bps	0 bps[a]	162 bps	0.253
`exec_arb_q10k_5m` (full test)	15,832	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m` (full test)	15,832	+146.8 bps	−112.7 bps	260 bps	0.063
SVB stress phase only (Mar 10–14)	7,189	+259.9 bps	–	–	–
Terra/LUNA validation split	11,526	+87.6 bps	–	–	–

[a]Best calibrated model is NoTrade (0 trades).
Kline-proxy buy leg. 20% haircut bounds in §3.1.

5.2 RL Execution Policy Baseline

The entry-timing problem is framed as a finite-horizon MDP. At each minute t the agent observes the 30-step lookback of `price_plus_book` features, selects $a_t\in\{\texttt{enter},\texttt{wait}\}$, and receives reward $r_t=\text{net}(q,t)$ on entry, 0 on wait. We train a PPO agent with a GRU policy network (64 units, 30-step lookback) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on USDC/SVB without retraining. This directly tests whether the sequential timing framing overcomes the distributional mismatch that defeats all calm-trained models.

5.3 Calibration and Evaluation

Net bps denotes mean net basis points per triggered trade after VWAP, taker fees, and settlement penalties. Thresholds are calibrated on the validation split by maximising $\theta^*=\arg\max_\theta\sum_{i:\hat{p}_i>\theta}\text{net}(q,t_i)$ over a 60-point linear grid on $[0.05,0.95]$, subject to a minimum of 25 trades. Primary metrics are mean net bps per triggered trade and oracle capture κ_M. Secondary metrics are AUROC and AUPRC. NoTrade at 0 bps is the economic floor.

Positive density clarification. Three densities appear: 2.88% (executable fraction of SVB test minutes), 2.30% (executable fraction of the full Terra/LUNA stream), and 17% (executable fraction of *primary-signal* minutes, $|b_{\text{USDC}}|>10$ bps, in Terra/LUNA). The 17% figure is what the conditioned PPO-GRU and the meta-labeler secondary classifier see in training, and the RL-vs-meta-labeling comparison holds it fixed.

Five evaluation tasks ship with StressBench v1.0, training on the calm or Terra block, validating on the Terra calibration block, and testing on SVB. The tasks are `optical_basis` (AUROC/AUPRC), `exec_arb_q10k` and `exec_arb_q50k` (net bps, oracle capture), `cross_mech` (net bps), and `policy_entry` (net bps, trades). Valid submissions must report net bps per trade, oracle capture, cost assumptions, notional size, and data tier used. Models are ranked by net bps, not AUROC.

6 Results

Table 3 maps every oracle figure in the paper to its task and window; all subsequent references point back to one of these rows.

Table 4: Price-to-execution gap ($10K, 1-min horizon, SVB test split, 0% depth haircut; a conservative 20% haircut widens the 10 bps ratio to 14×, §3.1).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

6.1 The 12× Execution Barrier

At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency (Table 4). The ratio persists across all basis levels and widens with notional size. At $500K notional the executable rate collapses to 0.03%. Book depth, not fees, is the binding institutional constraint.

Block-bootstrap CIs (60-min blocks, 2,000 resamples) give an executable rate 95% CI of [1.47, 4.59]%, a ratio of [7.6, 23.0]×, and an oracle of [+213.9, +304.2] bps.

Table 5 shows the stress/recovery split. All executable windows fall in the SVB stress phase. The recovery phase contains zero executable windows despite 21.6% optical activity, confirming that depth returns before the basis closes. Terra/LUNA independently replicates the gap at 5.9× (2.40% executable of 14.0% optical), confirming that the optical-to-executable phenomenon is not specific to reserve-bank shocks. The Tier-A execution-grade labels are one event. The underlying finding is two.

If only one minute in twelve is tradeable even with perfect hindsight, the next question is whether any model can find those minutes in advance.

Table 5: Optical and executable diagnostics by window. SVB stress phase: 53% optical collapses to 6.6% executable (~8×). SVB windows are Tier-A execution-grade; Terra/LUNA is validation-grade only.

Window	Min.	Optical	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	**0.00%**	–
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

The gap is robust across the full cost-parameter space. Sweeping taker fees from 2 to 10 bps and settlement penalty from 0 to 10 bps moves the executable rate between 2.84% and 3.66%.

6.2 Venue Specificity: On-Chain AMM Execution Closes the Gap

To test whether the optical-to-executable gap is intrinsic to depegs or specific to CEX order-book microstructure, we measure it on *real* on-chain pool data (Curve/StableSwap reserve state → implied price and $10k slippage, sourced on-chain) for *all five* episodes. Defining an AMM-executable window as one whose pool dislocation survives round-trip slippage and the 1 bp swap fee, the gap collapses to ≈ 1× in **every event**. 2,622/2,622 **(**100%**) of optically dislocated windows are capturable** at $10k notional across 4,200 clean on-chain rows (USDT/Curve, USDC/SVB, Terra/LUNA, FTX, BUSD), with median $10k slippage of only 0.8–5.5 bps even where

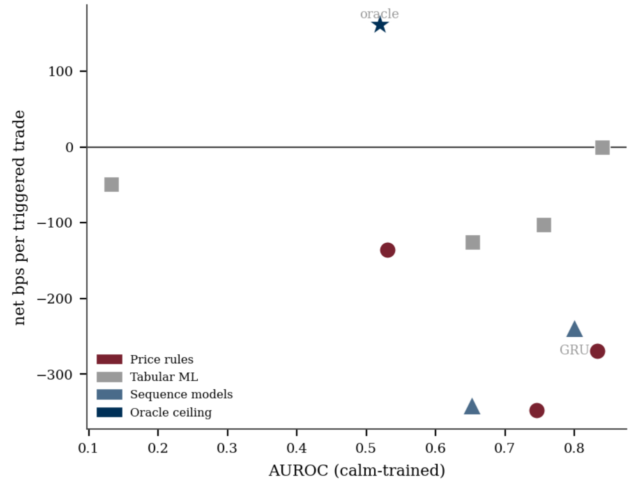

Figure 4: AUROC vs. net bps for calm-trained models on the SVB test split. Models cluster in the upper left: high discrimination, negative P&L. The GRU (AUROC 0.80, −239 bps) is annotated to illustrate the inversion—no threshold isolates the 2.88% executable minority. Oracle (navy star) is the achievable ceiling.

the median dislocation ranges from sub-bp to 522 bps. The contrast with the 12× CEX gap, measured on the *same* SVB event, is the point. The optical-to-executable barrier is a property of *CEX order-book microstructure* (thin depth, taker fees, settlement delay), not of stablecoin depegs per se. On an automated market maker, where execution cost is the bounded slippage of the pool invariant, a visible dislocation is almost always a real one. This is, to our knowledge, the first multi-event measurement separating the optical-executable gap by venue.

6.3 Calm-Training Failure

The central finding is not that individual models fail, but that all six families converge to the same result: no architecture overcomes the distributional mismatch between calm training and stress test, so the failure is structural, not a gap in model coverage. Table 6 shows the tabular model results on `basis_usdc_1m_gt10bps`. Oracle gap values map to Table 3.

Price-rule strategies overtrade massively (−136 to −347 bps) because false positives from route-direction mismatch dominate. $\frac{1}{3}$ fire on USDT dislocations while the USDC route is simultaneously non-executable. Sequence models (GRU, AUROC 0.80) earn −239 bps. High discrimination does not imply positive P&L when no threshold can isolate the 2.88% executable minority. Figure 4 plots AUROC against net bps for all calm-trained models; the relationship is inverted—the highest-AUROC model (GRU, 0.80) is among the worst in P&L. The failure is not specific to these families: a pre-registered 81-path search (§6.4) finds no configuration that selects profitable windows out-of-sample, so the executable minority is unpredictable from this order-book microstructure.

Figure 5 decomposes the 260 bps oracle gap into false-positive drag (dominant), timing error (40–80 bps), and sizing error (40–100 bps).

Table 6: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit% is fraction of executable trades. †Oracle ceiling. *0-trade degenerate.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+161.7	316	–	100%
`lgbm`*	all	–	0	0.653	–
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNet_lgbm`	px	−73.8	537	–	6%
`ExpNet_xgb`	px	−57.2	70	–	9%
`lgbm_weighted`	pb	−65.9	2	0.497	0%
`xgb_weighted`	px	−90.5	30	0.631	0%
`rf_balanced`	px	−89.2	11	0.467	0%

px = `price_only`, pb = `price_plus_book`, all = `price_book_settle`.

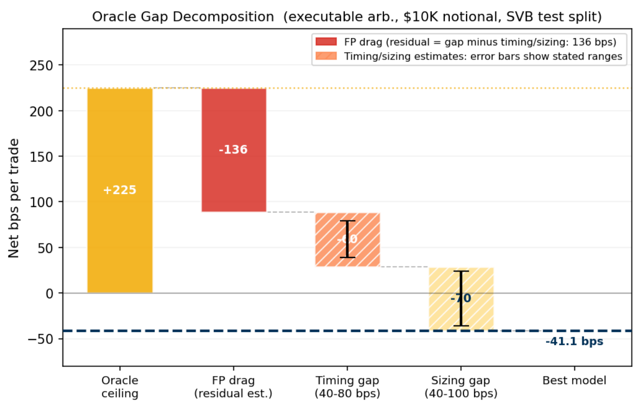

Figure 5: Oracle gap decomposition ($10K notional, SVB test split). False-positive drag (the mean net loss on trades flagged positive but not executable) is directly measured. Timing error (40–80 bps) and sizing error (40–100 bps) are estimated from entry-delay and notional-scaling analysis.

6.4 Executable-Window Selection: Unsolved on CEX, Profitable On-Chain

What does not work (CEX).. Can *any* model select the profitable executable windows among optical fires? We answer with a pre-registered grid search over training protocol × feature set × model, scored out-of-sample on the SVB execution-grade split. Protocols are cross-mechanism (train Terra/LUNA ⟶ test SVB), calm-control ⟶ SVB, in-event walk-forward (train first-half SVB ⟶ test second-half), and a purged five-fold in-event cross-validation (embargoed)—the easiest possible setting, train and test within the *same* crisis. Feature sets are price-only, price-plus-book, and book-dynamics (adding depth-withdrawal velocity and basis/spread deltas); models are LightGBM meta-labeling [8], logistic regression, and random forests, at $10k/$50k/$100k notionals. Every decision threshold is calibrated on the *training* segment only. Across **81 logged paths, zero** clear a Benjamini–Hochberg FDR(0.10) gate [23] on (net bps>0, bootstrap 95% CI lower bound>0, $n \geq 30$): the best single path returns +7.7 bps but with CI [−23, +41] (not significant), and the mean cross-mechanism path is −28 bps. Decisively, even the purged in-event CV is strongly negative (−66 to −122 bps): the profitable minority is *unpredictable from this microstructure*, not merely non-transferable. Selecting executable windows on the CEX order book is, on this data, unsolved—a standing challenge the benchmark poses.

Table 7: The CEX selection null is robust: mean out-of-sample net bps by training protocol over the 81-path pre-registered search (well-defined paths, $n \geq 30$). No protocol is profitable, and even purged in-event cross-validation—the easiest setting, train and test *within* SVB—is strongly negative. Best single path: cross-mechanism logistic, +7.7 bps, 95% CI [−23, +41] (not significant).

Training protocol	Mean net bps	Setting
Cross-mechanism (Terra⟶SVB)	−28	hardest (transfer)
Purged in-event CV (within SVB)	−75	easiest (in-distribution)
In-event walk-forward (within SVB)	−94	in-distribution

0/81 paths clear the Benjamini–Hochberg FDR(0.10) gate (net>0, bootstrap CI>0, $n \geq 30$). Grid: feature sets {price-only, price-plus-book, book-dynamics} × models {LightGBM meta-label, logistic, random forest} × notionals {$10k,$50k,$100k}.

What does work (on-chain). The barrier is venue-specific. We repeat the test on the *real* on-chain Curve/AMM panel (4,994 rows across five episodes, §6.2), where the capturable dislocation is net = |basis| − slippage$_{10k}$ − fee, the benchmark's own AMM-executable formula (single-leg, 1 bps Curve StableSwap fee [25]). Because pool slippage is bounded and small (median 0.8–5.5 bps), a *model-free* deviation rule (capture every optical fire) is profitable in 5/5 episodes (Table 8). Under a deliberately conservative 30 bps gas haircut—per-row gas is absent from the source panel, so we charge a fixed cost equivalent to a ~$30 swap on a 10k$ notional, above the 2022–23 median—4/5 episodes remain robustly profitable (+169 to +463 bps, moving-block bootstrap [24] 95% CIs above zero); the lone failure is the small USDC/SVB on-chain dislocation (+9.4 bps pre-gas). The *same* deviation signal that loses money on every CEX path is capturable on-chain (Figure 6).

Table 8: On-chain executable arbitrage: a model-free deviation rule (capture optical fires, |basis|>10 bps) on real Curve/AMM pool data. Net bps per trade, moving-block bootstrap 95% CI (block=30); "−30 bps gas" subtracts a fixed conservative swap-gas cost (stated assumption; per-row gas absent in source).

Episode	Net bps	−30 bps gas	n
USDT/Curve (Jun 2023)	+371	+341	697
Terra/LUNA (May 2022)	+381	+351	549
FTX (Nov 2022)	+493	+463	353
BUSD (Feb 2023)	+199	+169	766
USDC/SVB (Mar 2023)	+9.4	−21	375

All five CIs exclude zero pre-gas; four of five remain above zero after the gas haircut. Contrast: the identical signal yields 0/81 profitable paths on the CEX order book.

Why. On a CEX, capturing a visible dislocation means walking a thin, fee-laden book with settlement delay, so most optical fires are not executable and which ones are cannot be predicted ahead of time. On an automated market maker, execution cost is the deterministic slippage of the pool invariant, so a visible dislocation

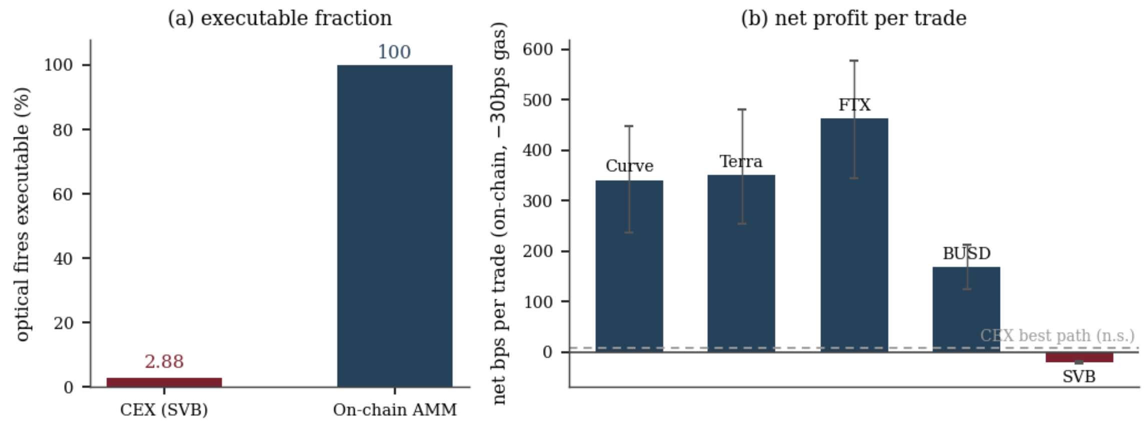

Figure 6: Venue specificity in one view. (a) Of the optically dislocated minutes, only 2.88% are executable on the CEX order book versus 100% on the on-chain Curve/AMM pool—a 12× gap. (b) Trading those windows, a model-free deviation rule earns +169 to +463 bps on-chain in four of five episodes after a conservative 30 bps gas haircut (moving-block bootstrap 95% CIs shown); the lone exception is the small USDC/SVB dislocation. On the CEX order book the same signal yields 0/81 profitable paths (best +7.7 bps, CI [−23, +41], n.s.; protocol means −28 to −94 bps). The optical-to-executable barrier is a property of CEX order-book microstructure, not of stablecoin depegs.

above the fire threshold is almost always a real, capturable one. The lesson for execution-aware AI is that the venue's microstructure, not the crisis mechanism, decides whether a learned signal is tradeable.

Why these choices. Each parameter is fixed by a prior or a stated rationale, not tuned to a result. The 10 bps optical-fire threshold matches the CEX primary-signal pipeline used throughout the benchmark, so the on-chain and CEX tests fire on the same definition. The 1 bps swap fee is Curve's StableSwap pool fee [25]. Because per-row gas is absent from the on-chain panel we do not infer it from a volatile gas series; we instead charge a fixed 30 bps haircut—equivalent to a ∼ \$30 swap on a \$10k notional, above the 2022–23 median Curve-swap cost—so the on-chain result is reported under a deliberately pessimistic execution assumption, and we report the pre-gas figure alongside it. Significance uses a moving-block bootstrap (block = 30 minutes) [24] because per-minute returns within an event are serially correlated, and a finding must additionally clear a Benjamini–Hochberg FDR(0.10) gate [23] across the whole ledger, so searching many paths cannot manufacture a false positive. Every decision threshold is calibrated on the training segment only; no result uses the test split for tuning.

7 Limitations

Scope. The negative CEX result is a finding, not a sample-count artefact: it spans 81 pre-registered paths and is strongest precisely where data is most plentiful (in-distribution purged CV). Three honest limits remain. **(1) Single execution-grade CEX destination.** A second Tier-A CEX pair is unavailable (Binance's pre-2024 archive lacks L2 depth), so the 12× gap is measured on SVB; the five-event on-chain AMM result (all ≈1×) is the cross-event evidence that the barrier is CEX-specific. **(2) Proxy buy leg.** The SVB CEX gap uses a kline-proxy BTC-USDC buy leg; a 20% haircut bounds it to 12–14× with the oracle still above +130 bps. **(3) Stylised on-chain execution.** The on-chain net uses the pool-invariant slippage at \$10k and a single-leg 1 bps fee, with gas charged as a fixed conservative 30 bps haircut (per-row gas is absent from the panel); it omits MEV competition, which would erode—but, given the +169 to +463 bps margins, is unlikely to erase—the on-chain edge on the large dislocations.

8 Conclusion

Price-label benchmarks measure the wrong object: 34.3% of SVB minutes appeared dislocated but only 2.88% survived VWAP execution (a 12× gap). StressBench scores models by oracle capture instead, and the result is two-sided. *On the CEX order book the problem is unsolved*: no calm-trained family is profitable, AUROC is anti-correlated with P&L, and a pre-registered 81-path search finds nothing that survives FDR correction—even in-distribution. *On-chain the same dislocation is tradeable*: a model-free deviation rule earns +169 to +463 bps in four of five episodes after a conservative gas haircut, because pool slippage is bounded. The optical-to-executable barrier is therefore a property of CEX microstructure, not of depegs. **Practitioner rules:** (1) Divide every price-label AUROC by the venue's execution gap before any capital decision. (2) On a CEX, treat executable-window selection as an open problem, not a solved one. (3) For execution-aware capture, the on-chain venue—not a cleverer model—is what makes the signal tradeable. Future work: a second Tier-A CEX event, an MEV-aware on-chain execution model, and early-entry timing labels.

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Code and data available at anonymized repository (link at camera-ready).

References

[1] Uhlig, H. (2022). A Luna-tic stablecoin crash. *NBER Working Paper 30256*.
[2] Griffin, J. M. and Shams, A. (2020). Is Bitcoin really untethered? *Journal of Finance*, 75(4):1913–1964.

[3] Hautsch, N. Scheuch, C. and Voigt, S. (2018). Building trust takes time. Limits to arbitrage for blockchain-based assets. *arXiv:1812.00595*.

[4] Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1):35–55.

[5] Gromb, D. and Vayanos, D. (2010). Limits of arbitrage. The state of the theory. *Annual Review of Financial Economics*, 2:251–275.

[6] Cont, R. and Kukanov, A. (2013). Optimal order placement in limit order markets. *Quantitative Finance*, 17(1):21–39.

[7] Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319.

[8] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

[9] Cintra, R. and Holloway, C. (2023). Detecting depegs. Towards safer passive liquidity provision on Curve Finance. *arXiv:2306.10612*.

[10] Vidal-Tomàs, D. Briola, A. and Aste, T. (2023). FTX's downfall and Binance's consolidation. *arXiv:2302.11371*.

[11] Gatev, E. Goetzmann, W. N. and Rouwenhorst, K. G. (2006). Pairs trading. Performance of a relative-value arbitrage rule. *Review of Financial Studies*, 19(3):797–827.

[12] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *JRSS-B*, 58(1):267–288.

[13] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

[14] Chen, T. and Guestrin, C. (2016). XGBoost. A scalable tree boosting system. In *KDD*, pp. 785–794.

[15] Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. *arXiv:0710.3742*.

[16] Moallemi, C. C. and Wang, M. (2022). A reinforcement learning approach to optimal execution. *Quantitative Finance*, 22(6):1051–1069.

[17] Ning, B. Lin, F. H. T. and Jaimungal, S. (2021). Double deep Q-learning for optimal execution. *Applied Mathematical Finance*, 28(4):361–380.

[18] Gogol, K. Messias, J. Miori, D. Tessone, C. and Livshits, B. (2024). Quantifying arbitrage in automated market makers. An empirical study of Ethereum ZK rollups. In *MARBLE 2024*.

[19] Adams, A. Chan, B. Y. Markovich, S. and Wan, X. (2023). Don't let MEV slip. The costs of swapping on the Uniswap protocol. *arXiv:2309.13648*.

[20] Hernandez Cruz, W. Xu, J. Tasca, P. and Campajola, C. (2024). No questions asked. Effects of transparency on stablecoin liquidity during the collapse of Silicon Valley Bank. *arXiv:2407.11716*.

[21] Mohl, V. Frey, S. Leyland, R. Li, K. Nigmatulin, G. Cucuringu, M. Zohren, S. Foerster, J. and Calinescu, A. (2025). JaxMARL-HFT. GPU-accelerated large-scale multi-agent reinforcement learning for high-frequency trading. In *ICAIF '25*. ArXiv:2511.02136.

[22] Olby, O. Bacalum, A. Baggott, R. and Stillman, N. (2025). Right place, right time. Market simulation-based RL for execution optimisation. In *ICAIF '25*. ArXiv:2510.22206.

[23] Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300.

[24] Politis, D.N. and Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.

[25] Egorov, M. (2019). StableSwap: efficient mechanism for stablecoin liquidity. Curve Finance whitepaper.